

SPSS

Statistical Package for Social Sciences

Introduction to Computer:-

A computer is an electronic device, operating under the control of instruction stored in its own memory that can accept data (input), process the data according to specified rules produce information (output) and store information for further use.

Introduction to windows:-

Windows is a multitasking operating system which use graphical user interface to interact with the user. It allows users to interact view, store files, run software, play games, watch video and provide a way to connect to the internet. It was released for both home computing and professional works. Microsoft introduced the first version

as 1.0 It was released for both home computing and professional function of window on 10 November 1983. Later it was released on many version of windows as well as current version.

Induction to SPSS:-

SPSS is a windows based program that can be used to perform data entry and analysis and to create table and graphs. SPSS is capable of handling large amount of data and can perform all of the analysis covered in the text and much more. SPSS is commonly used in the social sciences and in the business world so familiarity to this serve you well in the future. SPSS is updated often.

Opening SPSS:-

Depending on how the computer you are working on is structured. You can open SPSS in one of two ways:-

1- If there is an shortcut of SPSS on the desktop, simply put the cursor on it and double click the left mouse button.

2- Click the left mouse button on the start button on your screen, then put your cursor on all programs and left click the mouse select SPSS for window by clicking the left mouse button

- Use one of these approaches to open SPSS yourself.

Exiting SPSS:-

To closing SPSS, you can either left click on the close button ☒ locate on the upper right hand corner of the screen or select exit from the file menu.

Chose one of these approaches.

A dialog box appear for every open window asking you if you want to save it before exiting. Click on "NO" for

each dialog box if we don't want to save any file.

otherwise click on "Save" option and select the location where you want to save your file and then click "OK".

✓ Windows in SPSS:- imp

In SPSS environment It has eight different types of windows:

- Data editor
- Viewer
- Draft viewer
- Pivot table editor
- Chart editor
- Text output editor
- Syntax editor
- Script editor.

-: Data Editor:-

The data editor is a versatile spreadsheet like system for defining, entering, editing and displaying data. This window opens when you start an SPSS session and displays the contents of a data file. In

this window you can create new data files or modify existing data files. The data editor is like a spreadsheet where cases are represented in rows and variables are represented in columns.

-: Viewer / Draft viewer :-

The viewer makes it easy to browse your results, selectively show and hide output, change the display, order results, and move presentation-quality tables and charts between SPSS and other applications. This window opens automatically the first time you run a procedure that generates some output. The window displays all statistical results, tables and charts and allows you to edit the output and save it in an output file for later use.

The window is divided into two panes the left pane referred to as the outlines pane, contain an outline view of output contents and can be used to navigate through your output and control the output display which is much handy if you have a lot of output. The right

synagogue box



pane is the contents pane, and contains statistical tables, charts and text output.

If you select an item in the outline pane, the corresponding items in the contents pane is highlighted.

-: Pivot Table editor:-

Output displayed in pivot table editor can be modified in a number of ways. Using the pivot table editor, it is possible to edit text, rearrange rows, columns and layers add color, create multidimensional tables, and selectively hide and display results. You move into the pivot Table Editor from the viewer by selecting the table you want to edit and clicking the right mouse button to open the SPSS pivot table object.

-: Chart editor:-

High resolution, full color pie charts, bar charts, histograms, scatterplots, 3-D graphics and more are included as standard features in SPSS. These can be edited in the chart editor. Changes in colour, font, axes, rotations and chart types can also be made using the chart editor.

You move into the chart editor by viewer by selecting the chart you want to edit and clicking the right mouse button to open the SPSS chart object.

-: Text output editor :-

Text in the viewer that is not displayed in pivot tables or charts can be modify using the text output editor. Possible modifications include changes to font characteristics such as color, type, style, and size.

-: Syntax editor :-

Although most tasks can be accomplished by simple pointing and clicking the mouse, SPSS also provides as powerful commands language that allows you to save and automate many common tasks. The command language provides some functionality not found in the menus and dialogue boxes.

As you undertake particular procedure in SPSS, (you can paste your dialogue

box choice into a syntax window. This window contains the command syntax that reflects the choices you have made in selecting menu options. This commands can be edited to include special features not available through the pull-down menus and dialogue boxes, and they can be saved in a file for further use.

-: Script editor:-

Scripting and object linking and Embedding (OLE) automation allow you to customise and automate many tasks in SPSS. With the Script Editor, you are able to create and modify basic scripts within the program. At the top of each window you will have noticed a menu and an icon bar. We will now look at these particular features, which provide quick and easy access to the special features available in each of the SPSS windows.

Entering data:-

Following steps to enter data:-

- In the data view, click on the first cell in the SPSS Data Editor window. You will notice that a heavy border appears around the cell, indicating that the cell is now active.
- Type in the first value of id. This value is displayed in the cell editor at the top of the Data Editor window and also in the cell itself.
- Press enter or move to another cell by using the arrow keys or mouse. Data values are not recorded until you press Enter or select another cell. Remember that it is more efficient to code genders numerically.
- To move around your data file quickly, you can hold down the control key with an arrow key to take you to the limit of the file in that direction.

Defining a variable

The process of defining a variable has seven optional steps. The primary step is naming your variables and the other steps cover labels, missing variables, variable type, column format and measurement level.

-: Variable Names :-

Variable names must comply with certain rules:

- They must begin with a letter. The remaining characters can be a letter, any digit, a full stop or the symbols @, #, - or \$.
- They cannot end with a full stop or underscore.
- Blanks and special characters cannot be used (i.e.: !, ?, and *).
- They must be unique: duplication is not allowed.
- The length of the name cannot exceed 64 bytes.
- Reserved words cannot be

used as variable names (i.e.: TO, OR, AND)

- Names are not case sensitive; that is, they can be written in upper or lower case.

- Long variable names need to wrap to multiple lines in the output, so SPSS attempts to break lines at underscore and full stops and a change from lower to upper case.

-: Variable labels :-

Variable labels is the full discription of the variable name and is an optional means of improving the interpretability of the output. For example, the first variable you will name in the data file is (id) and the label for this variable is 'participant identification number'. Suggested variable labels for the other variables appear in the following table

Name	Label
Gender	optional
age	optional
Cost	amount spent

∴ Value Labels:-

It is possible to use alphanumeric codes for the variables: however, you may also wish to use a numeric code. For example, for gender you could assign a code of 1 for female and 2 for male. This type of variable is categorical because it has discrete categories. The variables allday and area are also categorical because it has discrete categories. When variables are measured using interval or ratio scales, coding is not relevant unless categorisation is required.

Name

Label

ID

not applicable

Gender

1 = Female

2 = Male

° allday

1 = use 24 hr

2 = not use 24 hr

cost

not applicable

age

not applicable.

∴ Missing Values :-

It is rare to obtain complete data sets for all cases. When dealing with missing data you may leave the cell blank or assign missing value codes. If you chose the latter, then a number of rules apply:

Missing value codes must be of the same data type as the data they represent. For example, for missing numeric data, missing value codes must also be numeric.

Missing value codes cannot occur as data in the data set.

By convention, the choice of digit is usually 9.

∴ Variable Type :-

By default, SPSS assumes that all new variables are numeric with two decimal places. However, it is possible to select other variable types and vary the number of decimal places.

-: Column Format :-

It is possible to adjust the width of the data editor columns and change the alignment of data in the column (left, center, right).

-: Measurement Level :-

You can specify the level of measurement as scale (interval or ratio) ordinal or nominal.

1- Working in the SPSS data editor window, double click a variable name at the top of the column in the data view or click the variable view tab.

-: Computing Frequencies:-

A frequency distribution is a display of the occurrence of each score value. The frequency distribution can be represented in tabular form or, with more visual clarity in graphical form. For continuous variables, measured on ratio or interval scales, histograms or frequency polygons are appropriate.

→ To compute frequency:-

- 1- Select the analyze menu.
- 2- Click on descriptive statistics and then on Frequencies to open the frequencies dialogue box.
- 3- Select the variables you require (i.e.: aces) and click on the ► button to move variable into variable box.
- 4- Click on the statistics command push button to open the Frequencies: Statistics sub-dialogue box.
- 5- In the percentiles value box, select the Quartiles check box.
- 6- In the central Tendency box, select the Mean, Median and Mode check box.

7- In the Dispersion box, select the std. deviation, Variance, Range, Minimum and Maximum check boxes.

8- Click on continue.

9- Click on the charts... command push button to open the Frequencies: Charts sub-dialogue box.

10- Click on the histograms radio button. You will notice that you can also obtain a normal curve overlay. So click on the With normal curve check box.

11- Click on Continue and then OK.

-: Computing New Variable :-

The compute command is most commonly used to obtain composite scores for items on the scales. This can be achieved by for the whole data set or for only a subset if certain conditions apply.

1- Select the transform menu.

2- Click on Compute to open the compute variable dialogue box

if previous settings remain, these can be cleared with the Reset command push button.

3- In the Target variable: box, type an appropriate variable name (i.e totalatt).

4- Select the first scale item from the source variable list (i.e: att1) and click ► button to move the variable into the Numeric Expression: box.

5- Click on the + button.

6- Select the second scale item from the source variable list (i.e: att2) and repeat step 4 and 5.

7- Click on OK.

-: Selection of Cases :-

In the select case option in the data menu, there are a number of procedures that can be chosen.

- Selection of specified cases using If option.

- Selection of Random cases using the Sample option.

- Selection of cases based on Time or range using Range option.

However, Pie chart can be difficult to read and interpret accurately.

→ Histogram:-

A histogram is a type of chart that displays the distribution of continuous data by dividing the data into intervals called "bins" and plotting the frequency or proportion of data points that falls into each bin. The bars in the histogram are usually adjacent to each other, and the width of each bar is proportional to the size of the corresponding bin.

→ One Sample t-test:-

The one sample t-test is used when you have data from a single sample of participants and you

wish to know whether the mean of the population from which the sample is drawn is the same as the hypothesised mean.

⇒ To conduct one sample t-test:-

- Select the 'Analyze' menu.
- Click on 'compare means' and then one sample t-test to open the one sample t-test dialogue box.
- Select the variable you require and click on the button to move the variable into the test variable.
- In the 'Test Value' box, type the mean score.
- Click on OK.

Recoding:-

You can modify data values by recoding. There are three instances where recoding variable is appropriate.

→ To collapse a continuous variable:-

- Select the Transform menu.

- Click on recode and

Into different variables to open the Recode into different variable dialogue box. By selecting Into different variables, you will retain the original data.

- In the source variable list, select the variable you wish to recode and click on the ► button to move the variable into the Input variable → output variable box.

- In the Name box of the Output variable box, type the new variable name. (i.e: agecat)

- Click on the change command push button.

- Click on the Old and new values... command push button to open the Recode into different variable: old and new variables sub-dialogue box.

- In the old value box. Click on the Range, Lowest through value: radio button and type the median in the box.

- In the new value box, click on the range new value.

click on add command.

- In old value box, click on the range, value through Highest: radio button and Type median.

- In new box type new value.

- Click on add command.

If you go back into your data file, you will notice that the variable

(agecat) now has two possible values 1 = 0-44 and 2 = 45+ years. This

are used in analysis where

categories are required.

Date: / /

→ To Recode negatively worded scale:-

Three of the items on the attitude scale required recoding because they are negatively worded. Remember that the response format for the items was 1 = strongly agree, 2 = agree, 3 = neutral, 4 = disagree, 5 = strongly disagree.

- Select the Transform menu.
- Click on Recode and Into same variables to open the Recode into same variables dialogue box. By selecting Into same variable, you will overwrite the original data.

- In the source variable list select the variables you wish to recode (i.e. att2, att4, att6) and click on ► button to move the variable into variable box.

- Click on the old and new values... command to open the Recode into same variable : old and new values sub dialogue box.

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- In old value box type old value
- In new value box type the new value
- Click on add command.
- In the old value box type second old value.
- In the new value box type the second new value.
- Click on add command.
- Repeat for remaining values.

-8 Missing Values:-

When dealing with the missing data you may leave the cell blank or assign missing value codes. Some rules are apply to assign missing code.

Missing value codes must be of the same data type as the data they represent. For example for missing numeric data, value code must be also numeric.

Missing value codes cannot occur as data in the data set.

By convention the code of digit is usually 9.

→ Replace missing values:-

Missing observation can be problematic. To avoid this problem, you can replace missing value with estimates computed using one of several methods. A commonly used method is mean substitution.

- Select the Transform menu.
- Click on recode into same variable to open same variable dialogue box.
- In the source variable list select the variables into you wish to recode and click on the ► button to move the variable into variables box.

- Click on the old and new values... command to open the recode into some variable: old and new values sub dialogue box

- In the old value box select the system-or-user-missing radio button.

- In the value new box, type the mean of the variable

- Click on the add command.

→ Exploring Variable

Exploring variable is the process of analyzing data to find patterns, relationship, and trends. This is done by statistical software to create graphs, charts that help visualize the data. Exploring variable can help researchers identify important factors that

may influence an outcome, such as the relationship between age and income. It can also help identify outliers, or data points that are significantly different from the rest of the data. By exploring variables, researchers can gain a better understanding of the data they are working with and make more informed decisions.

→ Time Series

A Time series is a set of data points collected over time and is used to track changes in a specific variable. Time series analysis is used in many fields.

→ Create Time series:-

To create Time series

in SPSS, you need to follow this steps.

- Open SpSS and create a new data set.
- Click on the "data viewer" tab on and enter data and time information in the first column.

Make sure the format is correct.

- Enter the data values in the second column.
- Click on the variable view tab and change the type of first column of data.
- Change the format of the first column to the desire format.
- Save the data set.

⇒ To analyze time series you can use the 'Time Series' module in SPSS. You can use this module to create forecasts, identify trends and others.

→ Defining Data Variable:-

Defining data variable is the process of assigning labels and values to different categories or variable in a data set. This is important because it allows researchers to analyze the data more easily and accurately.

For example: If you were conducting a survey on people's favorite colors, you would define the data variable by assigning a label to each color and a value to each response. You might assign the label Blue to the value "1", Red to "2",

Green to "3" and so on. This would allow you to easily analyze the data by color and determine the most popular color.

→ It is done in the "Variable View" in SPSS.

→ Defining weights variable:-

Defining weights variables is the process used in SPSS to adjust the contribution of each observation in the data set. This is done to account for difference in the probability of an observation being selected or to correct for biases in the data.

For example:

If you were conducting a survey and you wanted to ensure that each respondent has an equal chance of being selected, you might assign a weight variable to each observation. The weight variable would be proportional to the inverse of the sampling probability for each observation.

→ It is done in 'Variable View' tab in SPSS.

→ Finding descriptive Statistics:-

To find descriptive statistics in SPSS, you can follow these steps:

- Open the data set in SPSS.
- Click on "analyze" in the menu bar at the top of the screen.
- Click on "Descriptive Statistics" in the drop down menu.
- Click on descriptives.
- Select the variable you want to analyze from the list on the left side of screen and move them to the "Variable" box on the right side of the screen.
- Click on "Options" and select the statistics you want to compute, such as mean.
- Click continue to return to the "Descriptive" dialog box.
- Click "OK" to run analysis.

→ Cross tabulation:-

Cross tabulation in SPSS is a process of analyzing the relationship between two categorical variables.

To perform cross tabulation in SPSS, follow these steps. It involves creating a table that shows the frequency distribution of the variable for each category of the other variable.

For example:- If you were interested in analyzing the relationship between gender and political affiliation, you could use cross tabulation to create a table that shows number of male and female in each political party.

⇒ Measures of association:-

Measures of association is a statistical techniques used to determine the degree of relationship between two or more

variables. These measures are used to examine the strength and direction of the relationship between variables:

⇒ The common measures of associations are:-

1- Pearson's correlation coefficient:

This measure is used to determine the strength and direction of the linear relationship between two continuous variables. It ranges from -1 to $+1$ with values closer to -1 indicates negative correlation, closer to $+1$ indicates positive correlation, closer to 0 (zero) indicates no correlation.

2- Spearman Rank correlation coefficient:

Used to determine strength and direction of relationship between ordinal variable. Its range is same as above.

3- Phi coefficient: used to measure and determine the strength and direction of the relationship between two dichotomous variables.

4- Cramer's V: This is used to measure or determine the strength and direction of the relationship between two nominal variable ranges from 0 to 1 with value closer to 1 indicates stronger relationship between the variables.

→ Merging files:-

Merging files in SPSS is a process of combining two or more data sets into a single dataset. This is done when you have data that is spread across multiple files and you want to combine them into a single file for analysis.

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→ Methods of Merging files:-

1- Adding cases: This method is used when you have two database with the same variables and you want to combine them into single datasets. In this cases from second added to the end of first.

2- Adding Variables: This method is used to when you have two datasets with different variables and you want to combine them into a single datasets. In this variables from the second added at the end of first.

3- Matching cases: This method is used when you have two datasets of with a common variable and you want to combine

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them into a single dataset.
In this the cases from
second dataset matched to
the cases in first based on
common variable.

→ Splitting Files:-

Splitting files in SPSS is a process
of dividing a single dataset into
multiple datasets based on the
values of one or more variables.

This is done when you want to
analyze subsets of the data separately
or compare the results of different
analyses on different subsets of
data.

-: Method of Splitting:-

1. Split file: This method is used
to split a dataset into multiple
datasets based on the values of
one or more variables. In this

Method you specify the variable to split on, and SPSS creates a separate dataset for each unique value of variables.

2- Select cases:

This method is used to create a subset of dataset based on the values of one or more variables. In this method, you specify the condition of selecting cases, and SPSS creates a new dataset that contains only the cases that meet conditions.

3- Filter:

This method is used to create a temporary subset of a dataset based on the one or more values of variable. In this method, you specify the conditions for filtering cases, and SPSS hide that case that not meet conditions.

-: Bar chart :-

Bar chart is used to represent data visually in SPSS.

A bar chart displays data as a series of bars, where the height of each bar represents the frequency or proportion of the corresponding category. It is useful for comparing the frequency or proportion of different categories and for visualizing the distribution of categorical data.

-: Pie chart :-

A pie chart displays data as a circle divided into slices, where the size of each slice represents the proportion of the corresponding category. It is useful for showing the relative sizes of different categories and for visualizing the composition of a whole.

-: Selection using If :-

- 1- Select the Data menu.
- 2- Click on Select cases or click on the select cases tool to open the select cases dialogue box.
- 3- In the select box, click on the If condition is satisfied radio button.
- 4- Click on the If... command pushbutton to open the Select cases: If sub-dialogue box.
- 5- Select the variable you require and click on the ► button to move the variable into the box.
- 6- Click on the operator of your choice (i.e.: $\leq$) which will be pasted into the box above.
- 7- Type the value you required (i.e. 50).
- 8- Click on continue and OK.